Chapter 1

Creating webpage layout using HTML and AI support

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1.1 Web Technology

1.1.1 The World Wide Web -www

The World Wide Web (WWW) is a network of online content that is formatted in HTML and accessed via HTTP. The term refers to all the interlinked HTML pages that can be accessed over the Internet. The World Wide Web was originally designed in 1991 by Tim Berners-Lee while he was a contractor at CERN.

The World Wide Web is most often referred to simply as "the Web."

1.1.2 The Internet

The **Internet**, sometimes called simply "the Net," is a worldwide system of computer networks - a network of networks in which users at any one computer can, if they have permission, get information from any other computer (and sometimes talk directly to users at other computers).

Using the Internet, computers connect and communicate with each other, primarily using the TCP/IP (Transmission Control Protocol / Internet Protocol). Think of TCP/IP as a book of rules, a step-by-step guide that each computer uses to know how to talk to another computer.

ISPs (Internet service providers)

ISPs (Internet service providers), the companies that provide Internet service and connectivity, also follow these rules. The ISP provides a bridge between your computer and all the other computers in the world, which are all a part of the Internet. The ISP uses the TCP/IP protocols to make computer-to-computer connections possible and transmit data between them. When connected to an ISP, you are assigned an IP address, which is a unique address given to your computer or network and allows it to be found while on the Internet.

URL (Uniform Resource Locator)

Abbreviated as *URL*, a Uniform Resource Locator is a way of identifying the location of a file on the internet. They're what we use to open not only websites, but also to download images, videos, software programs, and other types of files that are hosted on a server.

Opening a *local* file on your computer is as simple as double-clicking it, but to

open files on *remote* computers, like web servers, we must use URLs so that our web browser knows where to look. For example, opening the HTML file that represents the web page explained below, is done by entering it into the navigation bar at the top of the browser you're using.

Uniform Resource Locators are most commonly abbreviated as *URLs* but they're also called *website addresses* when they refer to URLs that use the HTTP or HTTPS protocol.

URL is usually pronounced with each letter spoken individually (i.e. *u - r - l*, not *earl*). It used to be an abbreviation for *Universal Resource Locator* before being changed to Uniform Resource Locator.

Examples of URLs

You're probably used to entering in URL, like this one for accessing Google's website:

https://www.google.com

The entire address is called the URL. Another example is this website (first) and Microsoft's (second):

https://lifewire.com https://www.microsoft.com

1.1.2 Web Server and Hosting

Server hosting refers to the outsourcing of an organization's server placement and platform to a third-party Managed Hosting Provider (MSP). A client uses the Internet to connect to data and applications on a managed server and pays a recurring fee to the hosting provider. A MSP usually operates and manages large data centers with dozens, hundreds or thousands of hosted servers for two or more clients. This model is known as colocation or colocated hosting.

The server hosting model provides a best-of-all-worlds scenario by giving organizations access to server platforms to host their applications and data without shouldering attendant costs.

1.2 What Is HTML, HTML5, and CSS?



HTML (the Hypertext Markup Language) and CSS (Cascading Style Sheets) are two of the core technologies for building Web pages. HTML provides the *structure* of the page, CSS the (visual and aural) *layout,* for a variety of devices. Along with graphics and scripting, HTML and CSS are the basis of building Web pages and Web Applications.

HTML5 (HyperText Markup Language, version 5) is the latest major version of the standard markup language used to create and structure content on the web. It is maintained by the World Wide Web Consortium (W3C) and the WHATWG.

HTML5 introduced:

- **Semantic elements** (<header>, <footer>, <article>, etc.)
- Native **audio and video support**
- Better **form controls**
- Built-in **APIs** (e.g., Canvas, Geolocation, Local Storage)
- Improved support for modern web applications

What is CSS?

CSS is the language for describing the presentation of Web pages, including colors, layout, and fonts. It allows one to adapt the presentation to different types of devices, such as large screens, small screens, or printers. CSS is independent of HTML and can be used with any XML-based markup language. The separation of HTML from CSS makes it easier to maintain sites, share style sheets across pages, and tailor pages to different environments. This is referred to as the *separation of structure (or: content) from presentation.*

1.3 Text Editor

A **text editor** is program that allows you to open, view, and edit plain **text** files. Unlike word processors, **text editors** do not add formatting to **text**, instead focusing on **editing** functions for plain **text**. **Text editors** are **used by** a wide variety of people, for a wide variety of purposes.

Editing HTML and CSS code can be done without any specific tools. In fact, if you have a simple text editor, you are good to go. However, just because you can do something doesn't mean it is the best way to do it – and that applies to web development as well.

Web development IDE does all the things simple text editors do plus a number of more advanced stuff that you can't do with text editors. For instance, while an editor such as Sublime or Atom can be used as an HTML CSS JavaScript editor, they only allow you to write code.

Of course, they come with a bunch of convenient features such as syntax highlighting, customizable interfaces, and extensive navigation tools, you will need additional features to make a functional app. For example, you will need a debugger and a compiler.

1.4 Fundamental Elements and Concepts in HTML

1.4.1 Structure

Many webpages act like electronic versions of documents such as Newspapers.

Think about the stories you read in a newspaper: for each story, there will be a headline, some text, and possibly some images. If the article is a long piece, there may be subheadings that split the story into separate sections or quotes from those involved. Structure helps readers understand the stories in the newspaper.

Structure is a way to build a webpage to make it easier for the reader to understand what he is reading. HTML code describes the structure of webpages.

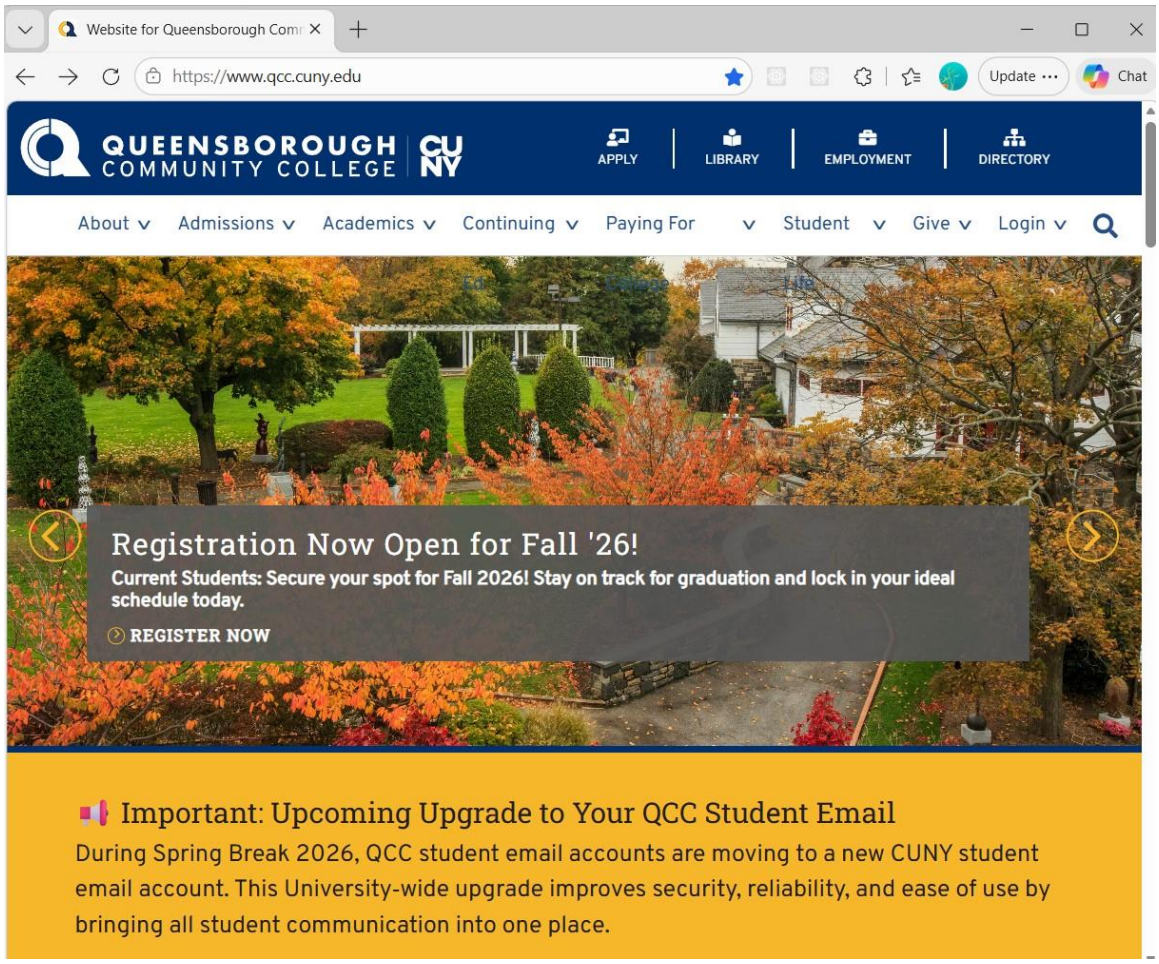


Figure 1 – Screenshot of the QCC website showing the website's structural template

1.4.2 The HTML code

The HTML code is made up of characters that live inside angled brackets which are called HTML **elements**. Elements are usually made up of two **tags:** an opening tag and a closing tag.



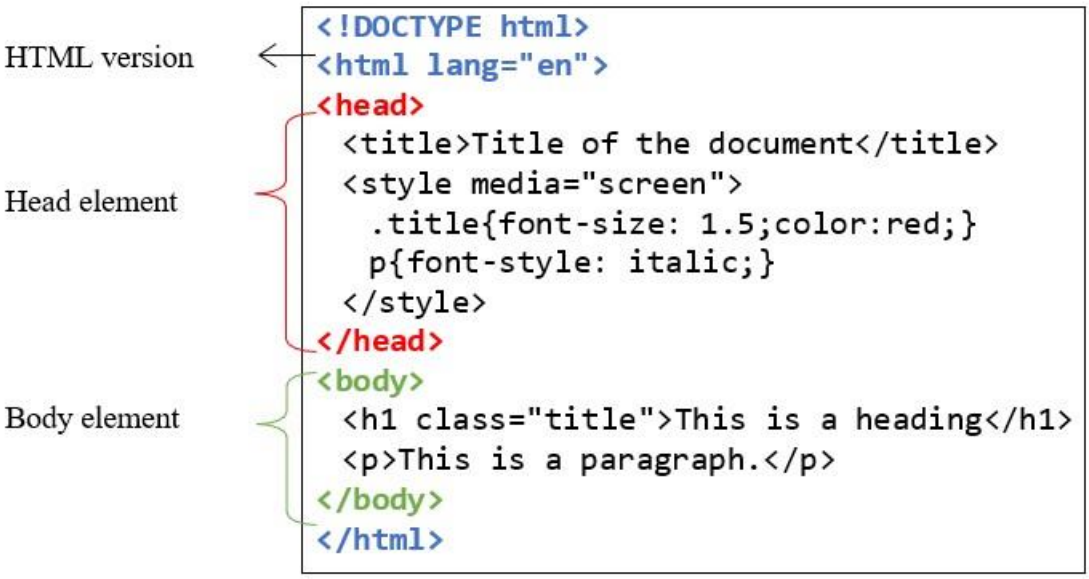
Each HTML element tells the browser something about the information that sits between its opening and closing tags.

HTML is built from **elements**, represented by **tags**.

1.4.3 Basic Structure of an HTML Document

An HTML document has three required parts:

- a line containing HTML version information. When the webpage is loaded, the declaration line is the first line that the browser reads. It tells the browser that the following file is an HTML file.
- a head element. The head element contains metadata, which is data about the HTML document, that is not displayed on the browser when the webpage is loaded, except for the title.
- a body element. Visible content in the web. The body element contains the document's actual content. Elements in the body build the visual part of a webpage.



1.5 Common HTML elements

1.5.1. Attributes

In HTML, **attributes** are extra pieces of information added to an element (tag) to modify its behavior, appearance, or provide additional details. An attribute is placed **inside the opening tag** and usually comes in this format:

```
<a href="https://example.com">Visit</a>
```

1.5.2. Text

```
<h1>Heading</h1>
<p>Paragraph</p>
<strong>Bold</strong>
<em>Italic</em>
```

1.5.3. Lists

Lists are commonly used in HTML to organize content in a clear and structured way. HTML provides three main types of lists: **ordered lists**, **unordered lists**, and **definition lists**.

Ordered Lists
Ordered lists are used when the sequence of items matters. Each item is numbered automatically. They are often used for steps in a process, instructions, or any content that follows a specific order.

The ordered list is created using the `<ol>` element.

Each item within the list is placed inside `<li>` (list item) tags.

Example:

```
<ol>
  <li>Never miss classes</li>
  <li>Never give up!</li>
</ol>
```

Exercise 1:
Open an AI website, create a new project folder, and name it **Web Development**. Inside this folder, enter the following prompt into the AI tool:

"Create an ordered list with three items. The items should be listed in alphabetical order."

With the result generated by AI, answer the following questions:

1. Is the result from the AI correct? Explain your answer.

2. What is different from the AI-generated result? Explain the differences.

Unordered Lists (Bullet Lists)
Unordered lists are used when the order of items does not matter. Items are displayed with bullet points instead of numbers.
The unordered list is created using the `<ul>` element.
Each item is placed inside `<li>` tags.

Example:

```
<ul>
  <li>Apples</li>
  <li>Oranges</li>
  <li>Grapes</li>
</ul>
```

In summary,

- `<ol>` is used for ordered (numbered) lists
- `<ul>` is used for unordered (bulleted) lists
- `<li>` defines each list item
- Lists help organize and present information clearly and logically

1.5.4. Using images on Web Pages

There are many reasons to include images on a website. Images can enhance the visual appeal and help communicate ideas more effectively. Common uses include adding logos, photographs, illustrations, diagrams, or charts.

To be effective, images should:

- Be relevant to the content
- Convey useful information
- Create the right mood or feeling
- Be easy to recognize quickly
- Match the website's color scheme

Stock Photos

If you don't have your own images, you can use stock photos or icons from online platforms. Some popular options include:

- www.pexels.com
- www.iconfinders.com

Choosing the Right Screen Size

Understanding screen sizes is important when designing a website so your images display properly across devices. Based on common screen resolutions, some widely used sizes include:

- 360 × 640
- 1366 × 768
- 1920 × 1080
- 375 × 667
- 1440 × 900
- 1280 × 800

Designing with these sizes in mind helps ensure your images look good on both mobile and desktop screens.

Image Formats

There are several common image formats used on websites:

- PNG (Portable Network Graphics)
- JPG/JPEG (Joint Photographic Experts Group)
- BMP (Bitmap)
- GIF (Graphics Interchange Format)
- SVG (Scalable Vector Graphics)

Understanding Images and Resolution

Images on a screen are made up of tiny squares called **pixels**. The resolution refers to the number of pixels displayed on the screen. Higher resolution generally means clearer and more detailed images.

When to use each format?

- **JPEG:** Best for photographs or images with many colors. It provides good quality with smaller file sizes.

- **PNG or GIF:** Best for images with fewer colors or large areas of solid color, such as logos, illustrations, and diagrams.

1.5.5. SVG (Scalable Vector Graphics)

SVG stands for **Scalable Vector Graphics**, a popular image format used on the web.

The `<image>` element in SVG allows you to include images inside SVG files. It can display both **raster images** (pixel-based images) and other SVG files. Raster images are made up of a grid of pixels and include formats like **JPEG, PNG, GIF, and ICO**.

SVG officially supports **JPEG, PNG, and SVG files**, while animated GIF behavior is not clearly defined.

Advantages of SVG

1. Text-Based Format

SVG files are written in text format, which improves accessibility. Search engines can read and index SVG content, making it easier to find online.

2. Scalability

SVG images can be resized without losing quality. They remain sharp on all screen sizes and devices.

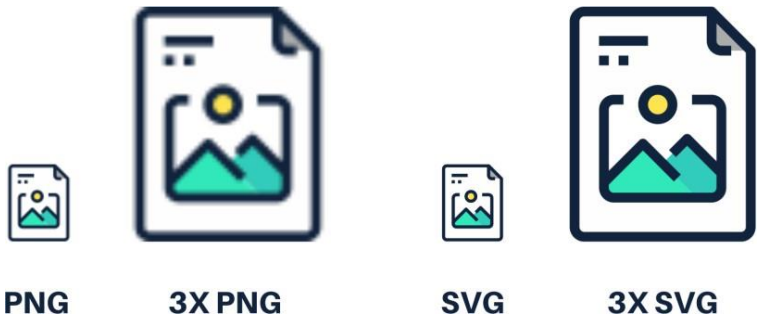


Figure 2 - Four images displaying a PNG and an SVG file enlarged to three times their original size.

You can also reuse a single SVG file in different sizes without creating multiple versions.

3. High Performance

SVG images can be embedded directly into HTML, reducing the need for extra file requests. This helps webpages load faster, improving performance and user experience.

4. Small File Size

SVG file size depends on elements like colors and shapes, not pixels. Unlike PNG or JPEG, increasing image size does not significantly increase file size.



Figure 3 - Two images—an SVG on the left and a PNG on the right—demonstrate that at the same image resolution, the SVG file size is approximately half that of the PNG file.

5. Easy Editing and Animation

SVG images can be edited using:

- Vector design software
- Text editors
- CSS (for colors and size)

They can also be animated using:

- CSS animations

- JavaScript
- Web animation tools

6. Integration with Web Technologies

SVG works well with **HTML, CSS, and JavaScript**, making it easy to use in modern web development.

7. W3C Support

The **World Wide Web Consortium (W3C)** developed and supports SVG, ensuring it remains a reliable web standard.

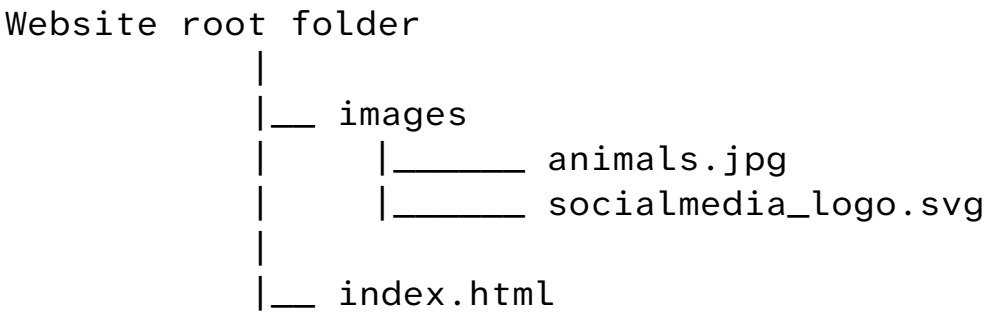
Disadvantages of SVG

- Complex images with many details can result in large file sizes
- Not suitable for detailed images like photographs
- Best used for simple graphics with fewer colors

Use SVG for logos, icons, and diagrams
Use JPEG or PNG for photos and detailed images

Storing Images

As a website grows, it is best to organize images in a separate folder, usually called **"images"** as shown in the folder/files organization diagram below:



For larger websites, you can create subfolders, such as:

- **interface** (logos, buttons)
- **products** (product images)
- **news** (news-related images)

Adding Images to a Webpage

To add an image in HTML, use the `<img>` tag:

```
<img src="image.jpg" alt="Description">
```

The `<img>` element is empty, meaning it does not have a closing tag.

Common <img> Attributes

- **src** → Specifies the image location (URL)
- **alt** → Describes the image (important for accessibility)
- **title** → Extra information shown as a tooltip

Exercise 2: In an AI prompt, type: *"Generate an HTML image element with a valid src attribute and an alt attribute set to 'QCC logo'."*

What code was generated?

1.5.6. Links in HTML

Links are one of the most important features of the web. They allow users to move from one webpage to another, making browsing or "surfing" the internet possible.

Creating Links

Links are created using the **<a> (anchor)** element. Anything placed between the opening **<a>** tag and closing **** tag becomes clickable.

Default Link Appearance

By default, links appear as:

- Blue and underlined (unvisited)
- Purple and underlined (visited)
- Red and underlined (active)

Common <a> Attributes

- **href** → Specifies the destination URL
- **target** → Specifies where the link opens

Common target values:

- _self → Opens in the same tab
- _blank → Opens in a new tab

Linking to External Websites

To link to another website, use the **href** attribute with the full URL.

```
<a href="https://www.qcc.cuny.edu" target="_blank"
rel="noopener noreferrer">
      Queensborough Community College
</a>
```

Why use rel="noopener noreferrer"?

- **noopener** → Prevents the new page from accessing your page (improves security)
- **noreferrer** → Hides the source of the link (improves privacy)

Email Links

To create an email link, use **mailto:** in the **href** attribute.

```
<p>
  Send an email to:
  <a href="mailto:profwu@college.edu">Professor Wu</a>
</p>
```

Linking Within the Same Page

To link to a specific part of a page:

1. Assign an id to an element
2. Link to that id using #

```
<img src="images/ny.jpg" id="bottom" alt="New York City">
<p><a href="#bottom">Go to Image</a></p>
```

Linking to a Section on Another Page

If the target is on another page, include the filename:

```
<p>
      <a href="index.html#bottom">Go to Another Page</a>
</p>
```

Linking Between Pages on the Same Site

Organize your website into folders. The main homepage is usually named **index.html**.

Examples

- **Same folder:**

```
<p>
   <a href="projects.html">Professional Projects</a>
</p>
```

- **Child folder:**

```
<p>
   <a href="images/imagesGallery.html">Gallery</a>
 </p>
```

- **Parent folder:**

```
<p>
   <a href="../index.html">Go back to homepage</a>
</p>
```

Key Takeaways

- Use `<a>` to create links
- Use `href` to define the destination
- Use `target="_blank"` to open links in a new tab
- Use `#id` to link within a page
- Use folder paths to link between pages

Exercise 3: in an AI prompt, type: *Generate HTML code with three linked social media icons (X, TikTok, Instagram) using images from the "images/logo" folder. Each icon should be wrapped in an* `<a>` *tag linking to its site and open in a new tab.*

What code was generated?

1.5.7 Creating Layout in HTML

HTML provides structure, while CSS is used to control layout and design.

Using Semantic Layout (Modern Approach)
Semantic HTML uses elements that clearly describe their meaning to both the browser and the developer.

Table 1: Common Semantic Elements and Their Purpose

Tag	Description
`<article>`	Independent, self-contained content
`<section>`	Thematic grouping of content
`<header>`	Introductory content or navigation
`<footer>`	Footer information for a page or section
`<nav>`	Main navigation links
`<aside>`	Sidebar or related content
`<main>`	Main content of the document
`<figure>`	Self-contained media (images, diagrams, etc.)
`<figcaption>`	Caption for a figure
`<details>`	Collapsible additional content
`<summary>`	Heading for <details>
`<time>`	Date or time
`<mark>`	Highlighted text

Key Semantic Elements Explained

`<section>`

- Represents a thematic grouping of content
- Often used for chapters, introductions, news, or contact sections

`<article>`

- Represents independent, self-contained content
- Can be reused or distributed separately

Examples:

- Blog posts
- Forum posts
- News articles
- Product cards

`<section>` and **`<article>`** can be nested inside each other depending on structure.

`<header>`

- Contains introductory content such as:

    - Headings (<h1>-<h6>)
    - Logos or icons
    - Navigation links

Note:

- Multiple <header> elements can exist in a page
- Cannot be placed inside <footer>, <address>, or another <header>

<footer>

- Contains closing or supporting information such as:
    - Copyright
    - Contact details
    - Sitemap
    - Related links
    - Back-to-top links

<nav>

- Defines major navigation links
- Not all links should be inside <nav>—only primary navigation menus
- Helps accessibility tools like screen readers

<aside>

- Represents content related indirectly to the main content
- Often used as a sidebar

<figure> and <figcaption>

- <figure> holds self-contained content like images or diagrams
- <figcaption> provides a caption for the figure

Example Page Layout

```
<body>
  <header>Header</header>
  <nav>Navigation</nav>
  <main>
    <section>Content</section>
    <aside>Sidebar</aside>
  </main>
  <footer>Footer</footer>
</body>
```

In summary:

- Block elements structure the layout and start on new lines
- Inline elements stay within text flow
- Semantic elements improve readability, accessibility, and organization
- HTML provides structure, while CSS handles styling and layout

1.6 How AI Can Help Create HTML Structure

AI can significantly speed up and simplify web development by helping developers generate, improve, and organize HTML structures efficiently.

1.6.1. Generate HTML from Description

You can describe what you want in plain English, and AI will generate the corresponding HTML structure.

Example Prompt:
Create a responsive landing page with a hero section and pricing cards (HTML code)

With the prompt, AI may generate:

- Semantic HTML structure
- Sections like `<header>, <main>, <section>`
- Responsive layout-ready markup
- Clean and organized code

An example of AI generated code might be as follows:

```
<header>
  <h1>Welcome to Our Website</h1>
</header>

<main>
  <section class="hero">
    <h2>Discover Our Services</h2>
    <p>High quality solutions for your needs.</p>
  </section>
```

```
  <section class="pricing">
    <div class="card">Basic Plan</div>
    <div class="card">Standard Plan</div>
    <div class="card">Premium Plan</div>
  </section>
</main>
```

1.6.2. Convert Design to HTML

AI tools can convert visual designs into code.
They can:

- Turn wireframes into HTML structure
- Convert Figma designs into HTML/CSS
- Generate code from images or mockups

Example Prompt:
Convert this layout description into HTML: a header with a logo, a navigation bar, and a main content section with two columns.

With the prompt, AI may generate:

```
<header>
  <img src="logo.png" alt="Logo">
  <nav>
    <a href="#">Home</a>
    <a href="#">About</a>
    <a href="#">Contact</a>
  </nav>
</header>

<main>
  <section class="content">
    <div class="left">Left Column</div>
    <div class="right">Right Column</div>
  </section>
</main>
```

1.6.3. Improve and Optimize Code

AI can review and enhance existing HTML code by:

- Refactoring -heavy code into semantic HTML5 elements
- Improving accessibility (e.g., adding alt, aria attributes)
- Suggesting SEO improvements
- Detecting and fixing validation errors

Example Prompt:
Improve this HTML by replacing unnecessary divs with semantic elements and adding accessibility features.

The given template is:

```
<!-- Header -->
<div>
     <div class="container nav-links">
       <h1>My Website</h1>
         <ul>
           <li><a href="#">Home</a></li>
           <li><a href="#">About</a></li>
           <li><a href="#">Services</a></li>
         </ul>
       </div>
       </div>
<!-- Main Content -->
<div>
   <!-- Slideshow -->
   <div>
      <div>
         <div >
            <img src="placeholder" alt="Slide 1">
         </div>
         <div >
            <img src="placeholder" alt="Slide 1">
         </div>
         <div >
            <img src="placeholder" alt="Slide 1">
         </div>
    </div>

    <!-- Introduction -->
    <div>
```

```
        <h2>Welcome to Our Website</h2>
        <p>
           This is a simple website template featuring a
           navigation header, an automatic slideshow, an
           introduction section, and a footer with copyright
           and social media links. You can customize it
           further based on your project needs.
        </p>
     </div>
</main>

<!-- Footer -->
<div>
  <div>
    <div class="copyright">
       <p>&copy; 2026 My Website. All Rights Reserved.</p>
    </div>
    <div>
       <p>Follow Us:</p>
       <a href="#">Twitter</a>
       <a href="#">Instagram</a>
    </div> </div>
</div>
```

1.6.4. Generate Boilerplate Faster

Instead of writing repetitive structures manually, AI can quickly generate common components such as:

- Forms
- Tables
- Navigation menus
- Dashboards
- Full webpage templates

Example Prompt:
Generate an HTML navigation menu with five links and a responsive structure.

With the prompt, AI may generate:

```
<nav>
  <ul>
    <li><a href="#">Home</a></li>
```

```
    <li><a href="#">Services</a></li>
    <li><a href="#">About</a></li>
    <li><a href="#">Blog</a></li>
    <li><a href="#">Contact</a></li>
  </ul>
</nav>
```

Indeed, AI helps developers:

- Quickly generate HTML from descriptions
- Convert designs into code
- Improve and clean existing HTML
- Create common layouts and components faster

By using well-structured prompts, developers can save time, reduce errors, and focus more on design and functionality rather than repetitive coding tasks.

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